

Qualification Pack



Assembly Process Sr. Technician - Wafer Thinning & Lapping

QP Code: TEL/Q7202

Version: 1.0

NSQF Level: 5

Telecom || 3rd Floor, Plot No 126, Sector - 44
Gurgaon - 122003 || email:QAteam@tsscindia.com

Qualification Pack

Contents

TEL/Q7202: Assembly Process Sr. Technician - Wafer Thinning & Lapping	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
TEL/N7205: Thinning and Lapping Processes for Telecom Applications	5
TEL/N7206: Operating and Maintaining Equipment for Telecom Wafer Processing	10
TEL/N7207: Quality Control and Safety in Telecom Semiconductor Manufacturing	15
DGT/VSQ/N0102: Employability Skills (60 Hours)	20
Assessment Guidelines and Weightage	27
<i>Assessment Guidelines</i>	27
<i>Assessment Weightage</i>	28
Acronyms	29
Glossary	30

Qualification Pack

TEL/Q7202: Assembly Process Sr. Technician - Wafer Thinning & Lapping

Brief Job Description

The individual for this job role is tasked with operating laser machines to accurately etch identification marks on semiconductor wafers. The individual is also responsible to set up laser parameters, ensuring the precision of markings, and maintaining equipment. In addition, he also need to inspect marked wafers for quality and adherence to specifications, documenting the process and outcomes.

Personal Attributes

The individual must demonstrate attention to detail, analytical and problem-solving abilities, and effective communication skills, both written and verbal. The individual should be adept at working collaboratively with others.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [TEL/N7205: Thinning and Lapping Processes for Telecom Applications](#)
2. [TEL/N7206: Operating and Maintaining Equipment for Telecom Wafer Processing](#)
3. [TEL/N7207: Quality Control and Safety in Telecom Semiconductor Manufacturing](#)
4. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Telecom
Sub-Sector	Semiconductor-Manufacturing & Packaging
Occupation	Semiconductor-M&P
Country	India
NSQF Level	5
Credits	18
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3522.9900

Qualification Pack

Minimum Educational Qualification & Experience	Graduate (2nd year of 3 year/ 4 year UG) with NA of experience OR 2-year Diploma after 12th grade (in any field) with NA of experience OR 12th Class (Electronics/Telecom/IT) with 3 Years of experience In relevant experience OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience relevant experience
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	28/02/2026
NSQC Approval Date	22/08/2024
Version	1.0
Reference code on NQR	QG-05-TL-02954-2024-V1-TSSC
NQR Version	1.0

Qualification Pack

TEL/N7205: Thinning and Lapping Processes for Telecom Applications

Description

This OS unit equips technicians with the skills to perform wafer thinning and lapping using appropriate techniques, ensuring consistent quality throughout the process.

Scope

The scope covers the following :

- Pre-Process Preparation
- Process Operation and Monitoring
- Post-Process Inspection and Documentation

Elements and Performance Criteria

Pre-Process Preparation

To be competent, the user/individual on the job must be able to:

- PC1.** interpret device specifications to determine target wafer thickness and surface finish
- PC2.** select the appropriate thinning and lapping based on wafer material properties and device needs
- PC3.** prepare the equipment for operation following established procedures (e.g., loading abrasive materials, preparing slurry solution, calibrating instruments)

Process Operation and Monitoring

To be competent, the user/individual on the job must be able to:

- PC4.** securely mount the wafer onto the processing platform of the equipment
- PC5.** configure and set up process parameters based on the technique and target specifications
- PC6.** operate the equipment, adhering to all safety protocols and Standard Operating Procedures (SOPs)
- PC7.** monitor the process using appropriate gauges and instruments to track critical parameters
- PC8.** analyze collected data to identify any deviations from desired specifications
- PC9.** recognize and troubleshoot minor equipment malfunctions during the thinning and lapping process, following established procedures
- PC10.** calibrate measurement instruments periodically to ensure accuracy of data collected during process monitoring

Post-Process Inspection and Documentation

To be competent, the user/individual on the job must be able to:

- PC11.** carry out minor adjustments to process parameters based on real-time monitoring to ensure consistent wafer quality
- PC12.** perform a final inspection of the wafer surface to identify any defects using appropriate tools and techniques
- PC13.** document the process parameters, results, and any observations in the designated logbook or electronic system

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relationship between wafer thickness and functionality of semiconductor devices
- KU2.** different surface finish requirements for various device applications
- KU3.** operating principles of different thinning and lapping techniques
- KU4.** impact of abrasive materials, slurry composition, and polishing pressure on thinning rate
- KU5.** influence of polishing pad type and polishing time on surface finish
- KU6.** functions and capabilities of equipment used in wafer thinning and lapping processes
- KU7.** safe handling procedures for wafers and hazardous materials used in the processes
- KU8.** common defects arising during thinning and lapping and their causes
- KU9.** safe mounting procedures for wafers onto processing platforms
- KU10.** standard operating procedures (SOPs) for setting up and configuring process parameters
- KU11.** functions and operating principles of measurement instruments used for monitoring critical parameters
- KU12.** potential safety hazards associated with wafer thinning and lapping processes
- KU13.** personal protective equipment (PPE) required for safe operation of equipment
- KU14.** techniques for interpreting data collected during process monitoring to identify trends and deviations
- KU15.** impact of process parameter variations on wafer quality (e.g., thickness uniformity, surface roughness)
- KU16.** principles of troubleshooting minor equipment malfunctions during the thinning and lapping process
- KU17.** techniques for performing final inspections of wafer surfaces to identify defects (e.g., visual inspection, microscopy)
- KU18.** importance of accurate and complete documentation of process parameters, results, and observations
- KU19.** record-keeping procedures for process data and quality control findings

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** read the relevant literature to get the latest updates about the field of work
- GS3.** listen attentively to understand the information/ instructions being shared
- GS4.** communicate politely and professionally
- GS5.** plan and prioritize tasks to ensure timely completion
- GS6.** coordinate with the co-workers to achieve the work objectives
- GS7.** evaluate all possible solutions to a problem to select the best one
- GS8.** take quick decisions to deal with workplace emergencies/ accidents

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Pre-Process Preparation</i>	8	15	-	3
PC1. interpret device specifications to determine target wafer thickness and surface finish	2	3	-	1
PC2. select the appropriate thinning and lapping based on wafer material properties and device needs	3	4	-	1
PC3. prepare the equipment for operation following established procedures (e.g., loading abrasive materials, preparing slurry solution, calibrating instruments)	3	8	-	1
<i>Process Operation and Monitoring</i>	14	30	-	4
PC4. securely mount the wafer onto the processing platform of the equipment	2	4	-	1
PC5. configure and set up process parameters based on the technique and target specifications	2	5	-	-
PC6. operate the equipment, adhering to all safety protocols and Standard Operating Procedures (SOPs)	2	3	-	-
PC7. monitor the process using appropriate gauges and instruments to track critical parameters	2	3	-	1
PC8. analyze collected data to identify any deviations from desired specifications	2	4	-	-
PC9. recognize and troubleshoot minor equipment malfunctions during the thinning and lapping process, following established procedures	2	5	-	1
PC10. calibrate measurement instruments periodically to ensure accuracy of data collected during process monitoring	2	6	-	1
<i>Post-Process Inspection and Documentation</i>	8	15	-	3

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. carry out minor adjustments to process parameters based on real-time monitoring to ensure consistent wafer quality	3	6	-	1
PC12. perform a final inspection of the wafer surface to identify any defects using appropriate tools and techniques	3	5	-	1
PC13. document the process parameters, results, and any observations in the designated logbook or electronic system	2	4	-	1
NOS Total	30	60	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N7205
NOS Name	Thinning and Lapping Processes for Telecom Applications
Sector	Telecom
Sub-Sector	
Occupation	Semiconductor-M&P
NSQF Level	5
Credits	6
Version	1.0
Last Reviewed Date	22/08/2024
Next Review Date	22/08/2025
NSQC Clearance Date	22/08/2024

Qualification Pack

TEL/N7206: Operating and Maintaining Equipment for Telecom Wafer Processing

Description

This OS unit equips technicians with the skills to operate wafer thinning and lapping equipment safely, perform routine maintenance, and troubleshoot basic operational issues.

Scope

The scope covers the following :

- Operate Thinning & Lapping Equipment Safely
- Perform Routine Equipment Maintenance
- Troubleshoot Basic Equipment Issues

Elements and Performance Criteria

Operate Thinning & Lapping Equipment Safely

To be competent, the user/individual on the job must be able to:

- PC1.** perform pre-operational checks on the equipment to ensure proper setup and functionality
- PC2.** wear and utilize appropriate PPE (e.g., gloves, safety glasses, respirators) based on the materials and processes involved
- PC3.** load and unload wafers onto the processing platform following safe handling procedures
- PC4.** start, operate, and monitor the equipment during the thinning and lapping process, ensuring adherence to established parameters
- PC5.** identify potential hazards associated with the operation of thinning and lapping equipment (e.g., moving parts, electrical hazards, chemical exposure)

Perform Routine Equipment Maintenance

To be competent, the user/individual on the job must be able to:

- PC6.** follow established maintenance schedules to perform cleaning tasks on the equipment (e.g., cleaning polishing pads, filters, waste disposal chambers)
- PC7.** lubricate moving parts according to manufacturer's recommendations
- PC8.** replace consumable items like abrasives, slurries, and filters as needed
- PC9.** inspect equipment components for signs of wear and tear (e.g., worn-out polishing pads, damaged belts)
- PC10.** report any identified equipment issues or potential maintenance needs to the designated personnel using appropriate documentation procedures
- PC11.** maintain accurate logs or records of completed maintenance tasks and identified equipment issues

Troubleshoot Basic Equipment Issues

To be competent, the user/individual on the job must be able to:

- PC12.** identify common operational problems encountered during the thinning and lapping process
- PC13.** analyze collected data to diagnose potential causes of the problem

Qualification Pack

- PC14.** attempt to troubleshoot the issue by following established procedures for minor adjustments
- PC15.** recognize limitations when troubleshooting complex issues and escalate problems to designated personnel for repair

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** functioning principles of wafer thinning and lapping equipment (e.g., mechanical grinding, CMP)
- KU2.** standard operating procedures (SOPs) for safe operation of specific thinning and lapping equipment models used in the workplace
- KU3.** potential hazards associated with the equipment (e.g., moving parts, electrical hazards, chemical exposure) and appropriate safety protocols
- KU4.** personal protective equipment (PPE) requirements for handling materials and operating the equipment safely
- KU5.** established preventive maintenance schedules and procedures for the specific equipment.
- KU6.** functions and purposes of different equipment components (e.g., polishing pads, filters, waste disposal systems)
- KU7.** signs of wear and tear on equipment components that may require replacement or repair.
- KU8.** importance of accurate record-keeping for completed maintenance tasks and identified equipment issues
- KU9.** common operational problems encountered during wafer thinning and lapping processes (e.g., slow thinning rate, uneven surface finish)
- KU10.** impact of process parameter variations on equipment performance (e.g., polishing pressure, slurry composition)
- KU11.** established procedures for troubleshooting and escalating complex equipment malfunctions.
- KU12.** limitations of basic troubleshooting and the importance of seeking assistance for complex issues

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work related records
- GS2.** read the relevant guides and literature to get the latest information about the field of work
- GS3.** communicate clearly and politely
- GS4.** listen attentively to understand the information/instructions being shared
- GS5.** plan and prioritize tasks to ensure timely completion
- GS6.** identify appropriate solutions to work related issues
- GS7.** take quick decisions in case of an emergency/accident

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operate Thinning & Lapping Equipment Safely</i>	12	26	-	4
PC1. perform pre-operational checks on the equipment to ensure proper setup and functionality	3	6	-	1
PC2. wear and utilize appropriate PPE (e.g., gloves, safety glasses, respirators) based on the materials and processes involved	2	4	-	-
PC3. load and unload wafers onto the processing platform following safe handling procedures	2	5	-	1
PC4. start, operate, and monitor the equipment during the thinning and lapping process, ensuring adherence to established parameters	3	6	-	1
PC5. identify potential hazards associated with the operation of thinning and lapping equipment (e.g., moving parts, electrical hazards, chemical exposure)	2	5	-	1
<i>Perform Routine Equipment Maintenance</i>	8	14	-	2
PC6. follow established maintenance schedules to perform cleaning tasks on the equipment (e.g., cleaning polishing pads, filters, waste disposal chambers)	1	2	-	-
PC7. lubricate moving parts according to manufacturer's recommendations	1	2	-	-
PC8. replace consumable items like abrasives, slurries, and filters as needed	1	3	-	1
PC9. inspect equipment components for signs of wear and tear (e.g., worn-out polishing pads, damaged belts)	2	2	-	-
PC10. report any identified equipment issues or potential maintenance needs to the designated personnel using appropriate documentation procedures	2	3	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. maintain accurate logs or records of completed maintenance tasks and identified equipment issues	1	2	-	-
<i>Troubleshoot Basic Equipment Issues</i>	10	20	-	4
PC12. identify common operational problems encountered during the thinning and lapping process	2	6	-	1
PC13. analyze collected data to diagnose potential causes of the problem	3	4	-	1
PC14. attempt to troubleshoot the issue by following established procedures for minor adjustments	3	6	-	1
PC15. recognize limitations when troubleshooting complex issues and escalate problems to designated personnel for repair	2	4	-	1
NOS Total	30	60	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N7206
NOS Name	Operating and Maintaining Equipment for Telecom Wafer Processing
Sector	Telecom
Sub-Sector	
Occupation	Semiconductor-M&P
NSQF Level	5
Credits	6
Version	1.0
Last Reviewed Date	22/08/2024
Next Review Date	22/08/2025
NSQC Clearance Date	22/08/2024

Qualification Pack

TEL/N7207: Quality Control and Safety in Telecom Semiconductor Manufacturing

Description

This OS unit equips technicians with the skills to perform quality control inspections of thinned and lapped wafers, analyze data, and ensure a safe work environment in the semiconductor assembly line.

Scope

The scope covers the following :

- Perform Quality Control Inspections
- Analyze Quality Control Data and Report Issues
- Maintain a Safe Work Environment

Elements and Performance Criteria

Perform Quality Control Inspections

To be competent, the user/individual on the job must be able to:

- PC1.** review quality control specifications for thinned and lapped wafer parameters
- PC2.** select appropriate inspection tools and measuring instruments based on the procedure
- PC3.** prepare a clean and well-lit work area for conducting visual inspections
- PC4.** perform visual inspections of thinned and lapped wafers to identify potential defects
- PC5.** utilize magnification tools for detailed inspection as needed
- PC6.** configure and calibrate measuring instruments according to established procedures (if applicable)
- PC7.** carefully measure relevant parameters of the thinned and lapped wafers
- PC8.** document identified defects and measured data using standardized formats

Analyze Quality Control Data and Report Issues

To be competent, the user/individual on the job must be able to:

- PC9.** compare collected data (visual inspection observations and measurement results) to quality control specifications
- PC10.** analyze trends in data to identify potential quality control issues
- PC11.** document any identified deviations from specifications along with the specific wafers affected
- PC12.** report quality control issues and deviations to designated personnel for further investigation and potential corrective actions

Maintain a Safe Work Environment

To be competent, the user/individual on the job must be able to:

- PC13.** attend and participate in mandatory safety training programs
- PC14.** maintain a clean and organized work area to minimize hazards
- PC15.** properly handle and dispose of hazardous materials according to SDS and waste disposal procedures

Qualification Pack

- PC16.** follow labeling and storage procedures for chemicals and materials
- PC17.** be aware of emergency exits, eyewash stations, and fire extinguishers
- PC18.** identify required PPE for different tasks and materials
- PC19.** select and wear appropriate PPE in good working condition
- PC20.** inspect PPE for damage before each use and report any issues
- PC21.** properly remove and dispose of PPE after use
- PC22.** respond promptly and safely to emergencies following established procedures

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** quality control specifications for thinned and lapped wafers (thickness, uniformity, roughness)
- KU2.** functions and capabilities of inspection tools and measuring instruments (microscopes, gauges, profilometers)
- KU3.** calibration procedures for measuring instruments (if applicable)
- KU4.** potential defects in thinned and lapped wafers and visual identification techniques
- KU5.** techniques for analyzing quality control data (trends, deviations from specifications)
- KU6.** impact of process parameter variations on final wafer quality
- KU7.** importance of accurate and timely reporting of quality control issues
- KU8.** documentation procedures for reporting data and deviations
- KU9.** safety regulations and protocols for the semiconductor assembly line
- KU10.** safe handling procedures for hazardous materials (chemicals, abrasives)
- KU11.** purpose and proper use of personal protective equipment (PPE) for different tasks
- KU12.** emergency response procedures for various scenarios (spills, fire, malfunctions)
- KU13.** location and function of safety equipment (eyewash stations, fire extinguishers)

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work related records
- GS2.** read the relevant guides and literature to get the latest information about the field of work
- GS3.** communicate clearly and politely
- GS4.** listen attentively to understand the information/instructions being shared
- GS5.** plan and prioritize tasks to ensure timely completion
- GS6.** identify appropriate solutions to work related issues
- GS7.** take quick decisions in case of an emergency/accident

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform Quality Control Inspections</i>	10	32	-	4
PC1. review quality control specifications for thinned and lapped wafer parameters	2	4	-	1
PC2. select appropriate inspection tools and measuring instruments based on the procedure	1	3	-	1
PC3. prepare a clean and well-lit work area for conducting visual inspections	1	3	-	-
PC4. perform visual inspections of thinned and lapped wafers to identify potential defects	1	4	-	-
PC5. utilize magnification tools for detailed inspection as needed	1	3	-	-
PC6. configure and calibrate measuring instruments according to established procedures (if applicable)	1	5	-	1
PC7. carefully measure relevant parameters of the thinned and lapped wafers	2	6	-	1
PC8. document identified defects and measured data using standardized formats	1	4	-	-
<i>Analyze Quality Control Data and Report Issues</i>	10	18	-	3
PC9. compare collected data (visual inspection observations and measurement results) to quality control specifications	2	4	-	1
PC10. analyze trends in data to identify potential quality control issues	3	6	-	1
PC11. document any identified deviations from specifications along with the specific wafers affected	2	3	-	-
PC12. report quality control issues and deviations to designated personnel for further investigation and potential corrective actions	3	5	-	1
<i>Maintain a Safe Work Environment</i>	10	10	-	3

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. attend and participate in mandatory safety training programs	1	1	-	-
PC14. maintain a clean and organized work area to minimize hazards	1	1	-	-
PC15. properly handle and dispose of hazardous materials according to SDS and waste disposal procedures	1	1	-	1
PC16. follow labeling and storage procedures for chemicals and materials	1	1	-	-
PC17. be aware of emergency exits, eyewash stations, and fire extinguishers	1	1	-	-
PC18. identify required PPE for different tasks and materials	1	1	-	1
PC19. select and wear appropriate PPE in good working condition	1	1	-	-
PC20. inspect PPE for damage before each use and report any issues	1	1	-	-
PC21. properly remove and dispose of PPE after use	1	1	-	-
PC22. respond promptly and safely to emergencies following established procedures	1	1	-	1
NOS Total	30	60	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N7207
NOS Name	Quality Control and Safety in Telecom Semiconductor Manufacturing
Sector	Telecom
Sub-Sector	
Occupation	Semiconductor-M&P
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	22/08/2024
Next Review Date	22/08/2025
NSQC Clearance Date	22/08/2024

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).

Qualification Pack

5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.

6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass 70 % aggregate for the QP.

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TEL/N7205.Thinning and Lapping Processes for Telecom Applications	30	60	-	10	100	30
TEL/N7206.Operating and Maintaining Equipment for Telecom Wafer Processing	30	60	-	10	100	30
TEL/N7207.Quality Control and Safety in Telecom Semiconductor Manufacturing	30	60	-	10	100	30
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	110	210	-	30	350	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.